

CHIRANJEEVI T N

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SOFTWARE VALIDATION ENGINEER

Seasoned Intel Validation Engineer with 10+ years of expertise in Xeon post-silicon validation, firmware qualification, silicon debug, Linux enablement, and HPC platform validation, driving end-to-end quality, root-cause analysis, and release readiness across multiple generations of Intel server platforms through deep technical leadership and automation-driven validation excellence.

ACCOMPLISHMENTS

- Received multiple Intel recognitions for post silicon validation excellence, including designation as DRA for GPU validation on Aurora and for debug patch qualification.
- Presented at-scale debug validation techniques at Intel's DCDC event, promoting scalable telemetry and automation across validation teams.
- Led cross-functional root cause analysis and release readiness for firmware/PCODE/BIOS patches, enabling timely production releases and customer-issue closure.

AREAS OF EXPERTISE

Xeon Post-Silicon & PCODE: Xeon platform from Gen4 to Gen6 | PCODE, OCODE, UCODE, RAPL, Crashlog, OSPL, IFWI/BIOS, ME | Power Management, Telemetry, Firmware Patch Qualification, Customer Issue Debug, Release | Readiness | Data Center GPU (ATS / PVC) : IFWI, PCode, AMC, CSC FW, HBMIO, PSC, PCIe Gen5, Xe-Link, RC6 | Media Transcoding, AI/ML Inference | firmware release qualification and Aurora platform readiness. | HPC Workloads: HPL, DGEMM, LAMMS, HACC, Nekbone, QMCPACK, GROMACS, GRID-QCD, HOTQCD, BQCD, NWChem, E3SM. | Android IVI & Automotive | Audio HAL, AudioFlinger, Audio Policy, Audio Server, DSP Integration, USB/HDMI Audio, VoIP, A2DP Offload, VTS Compliance; eAVB/TSN, IEEE 802.1AS gPTP, IEEE 1722 AVTP, SRP, Wireshark Packet Analysis Server & Linux | Clear Linux OS, Intel-next Kernel, BKC Validation, VT-d, Virtualization, Storage, Boot Validation, Cluster Qualification | Automation & Tools: PythonSV, PMX, EMON, ISD, TraceHub | Kubernetes, Docker, Jenkins, Airflow, OpenStack, ELK, Kafka, Prometheus, Grafana | OneBKC. | Testing & QA: Manual & Automation Testing, Selenium, SDLC/STLC, Test Planning, Test Case Design, Defect Tracking (Jira/Mantis), Regression, Functional, Integration, System and Performance Testing. | Leadership & Operations: Managed CW onboarding, execution tracking, and resource coordination to support large-scale validation programmes

PROFESSIONAL EXPERIENCE

INTEL, BANGALORE

December 2016 - August 2026

Software Validation Engineer

Android IVI Audio

- Validated Android IVI audio and multimedia subsystems including Audio HAL, AudioFlinger, Audio Policy, DSP integration, USB/HDMI Audio, VoIP, and VTS compliance.
- Led eAVB/TSN validation covering gPTP, AVTP, clock synchronization, and automotive audio-over-Ethernet streaming.
- Performed Wireshark-based packet analysis and root-cause debugging to ensure audio stream integrity, synchronization, and low-latency performance.

ATS/PVC GPU/Aurora

- Validated Intel Arctic Sound (ATS) Data Center GPUs across firmware, drivers, PCIe, virtualization, power management, and multi-card server platforms, enabling product bring-up and deployment readiness.
- Qualified media transcoding, AI/ML inference, and HPC workloads while leading debug of GPU, firmware, and platform issues to improve reliability, performance, and scalability of Intel's first-generation data centre GPU platform.
- Led validation of Intel Ponte Vecchio (PVC) GPUs, qualifying firmware releases, PCIe Gen5, Xe-Link, RC6, boot stability, and Aurora platform readiness.
- Executed HPC performance and regression validation across multi-GPU scale-up/scale-out systems, driving debug of firmware, GPU, and interconnect issues to improve platform reliability and scalability.
- Validated Aurora platform functionality, firmware releases, PCIe Gen5 compliance, GPU scalability, and system stability across large-scale HPC environments.
- Executed and debugged HPC/AI workloads including HPL, DGEMM, NAMD, GROMACS, GRID-QCD, HOTQCD, NWChem, and E3SM to identify performance and reliability issues.
- Drove platform readiness through firmware qualification, boot/reset validation, cluster triage, and cross-functional debug of hardware, firmware, driver, and workload issues.

Validation At-Scale

- Supported large-scale Intel Xeon platform validation environments, utilizing Kubernetes, Docker, Jenkins, Airflow, OpenStack, ELK, Kafka, Prometheus, and Grafana for automated test execution and telemetry monitoring.
- Managed and monitored 400+ validation nodes, enabling reliability validation, burn-in testing, BKC qualification, DPMO tracking, and platform readiness assessments.
- Analysed serial, BMC, Linux kernel, and hardware logs to identify root causes, debug system issues, and drive resolution in collaboration with cross-functional teams.

Xeon / PCODE

- Led post-silicon validation and release qualification of PCODE, OCODE, UCODE, RAPL, Crashlog, OSPL, IFWI/BIOS, and ME across Intel Xeon platforms.
- Executed firmware and microcode qualification, upgrade/rollback validation, and release sign-off across multiple operating system environments.
- Performed deep silicon debug using PythonSV, PMX, VISA, crash log, and platform telemetry to drive root-cause identification and issue resolution.
- Drove customer issue investigations and cross-functional debug efforts, improving platform stability, firmware quality, and release readiness.

EDUCATION

Master of Technology in Computer Science

Sambhram Institute of Technology, Bengaluru, VTU

Bachelor of Engineering in Computer Science & Engineering

U.B.D.T College of Engineering, Davangere, Davangere

Software Testing Training in Software Testing

QSpiders, Bengaluru

Manual Testing, Selenium Automation, SDLC, STLC, Test Planning, Test Case Design, Defect Tracking, Regression Testing, Functional Testing, Integration Testing.